

SKILL

Topic 2 to 10

```
varshini@DESKTOP-BLT5L3R:~$ ls
OSSP fork_exec fork_exec.c hello hello.c process.c
varshini@DESKTOP-BLT5L3R:~$ ls -l
total 52
drwxr-xr-x 11 varshini varshini 4096 Aug  2 14:58 OSSP
-rwxr-xr-x  1 varshini varshini 16400 Aug  3 06:04 fork_exec
-rw-r--r--  1 varshini varshini  946 Aug  3 06:04 fork_exec.c
-rwxr-xr-x  1 varshini varshini 15960 Aug  3 04:57 hello
-rw-r--r--  1 varshini varshini   81 Aug  3 05:22 hello.c
-rw-r--r--  1 varshini varshini   82 Aug  2 14:51 process.c
varshini@DESKTOP-BLT5L3R:~$ ls -a
.          .bashrc    .local          OSSP          hello.c
..         .cache     .motd_shown     fork_exec     process.c
.bash_history .config    .profile        fork_exec.c
.bash_logout .lessht    .sudo_as_admin_successful hello
varshini@DESKTOP-BLT5L3R:~$ |
```

```
varshini@DESKTOP-BLT5L3R:~$ cd
varshini@DESKTOP-BLT5L3R:~$ cd ..
varshini@DESKTOP-BLT5L3R:/home$ mkdir OSLab
mkdir: cannot create directory 'OSLab': Permission denied
varshini@DESKTOP-BLT5L3R:/home$ cd ~
varshini@DESKTOP-BLT5L3R:~$ pwd
/home/varshini
varshini@DESKTOP-BLT5L3R:~$ mkdir OSLab
varshini@DESKTOP-BLT5L3R:~$ ls
OSLab OSSP fork_exec fork_exec.c hello hello.c process.c
```

```

$ sudo apt install nano make
varshini@DESKTOP-BLT5L3R:/home$ cd ~
rmdir OSLab
varshini@DESKTOP-BLT5L3R:~$ find
.
./sudo_as_admin_successful
./bash_history
./bash_logout
./motd_shown
./fork_exec.c
./hello.c
./hello
./fork_exec
./lessht
./process.c
./config
./config/procps
./local
./local/share
./local/share/nano
./bashrc
./profile
./OSSP
./OSSP/README.md
./OSSP/.git
./OSSP/.git/hooks
./OSSP/.git/hooks/push-to-checkout.sample
./OSSP/.git/hooks/fsmonitor-watchman.sample
./OSSP/.git/hooks/post-update.sample
./OSSP/.git/hooks/pre-rebase.sample
./OSSP/.git/hooks/update.sample
./OSSP/.git/hooks/pre-push.sample
./OSSP/.git/hooks/applypatch-msg.sample
./OSSP/.git/hooks/sendemail-validate.sample
./OSSP/.git/hooks/pre-commit.sample
./OSSP/.git/hooks/prepare-commit-msg.sample
./OSSP/.git/hooks/pre-merge-commit.sample
./OSSP/.git/hooks/pre-receive.sample
./OSSP/.git/hooks/pre-applypatch.sample
./OSSP/.git/hooks/commit-msg.sample
./OSSP/.git/branches
./OSSP/.git/objects
./OSSP/.git/objects/pack
./OSSP/.git/objects/info
./OSSP/.git/info
./OSSP/.git/info/exclude
./OSSP/.git/HEAD
./OSSP/.git/refs
./OSSP/.git/refs/heads
./OSSP/.git/refs/tags
./OSSP/.git/config
./OSSP/.git/description
./OSSP/scripts
./OSSP/scripts/run.sh
./OSSP/assets
./OSSP/assets/architecture.png
./OSSP/bin
./OSSP/bin/my_shell
./OSSP/tests
./OSSP/tests/test_executor.c
./OSSP/tests/test_parser.c
./OSSP/fork_exec.c
./OSSP/LICENSE
./OSSP/obj
./OSSP/src
./OSSP/src/parser.c
./OSSP/src/executor.c
./OSSP/src/shell.c
./OSSP/src/builtin.c
./OSSP/src/main.c
./OSSP/fork_exec.c.save
./OSSP/.gitignore
./OSSP/include
./OSSP/include/parser.h
./OSSP/include/executor.h
./OSSP/include/shell.h
./OSSP/include/builtin.h
./OSSP/fork_exec
./OSSP/Makefile
./OSSP/docs
./OSSP/docs/design.md
./OSSP/docs/report.pdf
./cache
./cache/motd.legal-displayed
varshini@DESKTOP-BLT5L3R:~$ |

```

```

varshini@DESKTOP-BLT5L3R:/home$ locate hello.c
/home/varshini/hello.c
varshini@DESKTOP-BLT5L3R:/home$ touch
touch: missing file operand
Try 'touch --help' for more information.
varshini@DESKTOP-BLT5L3R:/home$ touch file1.txt
touch: cannot touch 'file1.txt': Permission denied
varshini@DESKTOP-BLT5L3R:/home$ locate hello.c
/home/varshini/hello.c
varshini@DESKTOP-BLT5L3R:/home$ cd ~
pwd
/home/varshini
varshini@DESKTOP-BLT5L3R:~$ touch file1.txt
ls
OSSP file1.txt fork_exec fork_exec.c hello hello.c process.c
varshini@DESKTOP-BLT5L3R:~$ cat file1.txt
varshini@DESKTOP-BLT5L3R:~$ echo "Hello, Linux!" > file1.txt
varshini@DESKTOP-BLT5L3R:~$ cat file1.txt
Hello, Linux!
varshini@DESKTOP-BLT5L3R:~$ cp file1.txt file2.txt
varshini@DESKTOP-BLT5L3R:~$ cp file1.txt /tmp/
varshini@DESKTOP-BLT5L3R:~$ cp file1.txt backup.txt
varshini@DESKTOP-BLT5L3R:~$ ls
OSSP file1.txt fork_exec hello process.c
backup.txt file2.txt fork_exec.c hello.c
varshini@DESKTOP-BLT5L3R:~$ cp file1.txt file2.txt

```

```

varshini@DESKTOP-BLT5L3R:~$ ls
OSSP fork_exec hello myfolder
backup_file.txt fork_exec.c hello.c process.c
varshini@DESKTOP-BLT5L3R:~$ mv backup_file.txt myfolder/
varshini@DESKTOP-BLT5L3R:~$ ls myfolder
backup_file.txt new.txt
varshini@DESKTOP-BLT5L3R:~$ mv backup_file.txt new.txt
mv: cannot stat 'backup_file.txt': No such file or directory
varshini@DESKTOP-BLT5L3R:~$ ls
OSSP fork_exec fork_exec.c hello hello.c myfolder process.c
varshini@DESKTOP-BLT5L3R:~$ ls myfolder
backup_file.txt new.txt
varshini@DESKTOP-BLT5L3R:~$ rm file.txt
rm: cannot remove 'file.txt': No such file or directory
varshini@DESKTOP-BLT5L3R:~$ ls
OSSP fork_exec fork_exec.c hello hello.c myfolder process.c
varshini@DESKTOP-BLT5L3R:~$ rm hello
varshini@DESKTOP-BLT5L3R:~$ ls
OSSP fork_exec fork_exec.c hello.c myfolder process.c
varshini@DESKTOP-BLT5L3R:~$ ls myfolder
backup_file.txt new.txt
varshini@DESKTOP-BLT5L3R:~$ rm myfolder/new.txt
varshini@DESKTOP-BLT5L3R:~$ ls
OSSP fork_exec fork_exec.c hello.c myfolder process.c
varshini@DESKTOP-BLT5L3R:~$ |

```

```
varshini@DESKTOP-BLT5L3R:~$ head hello.c
#include <stdio.h>

int main()
{
    printf("Hello, World!\n");

    return 0;
}
varshini@DESKTOP-BLT5L3R:~$ tail hello.c
#include <stdio.h>

int main()
{
    printf("Hello, World!\n");

    return 0;
}
varshini@DESKTOP-BLT5L3R:~$ more file.txt
more: cannot open file.txt: No such file or directory
varshini@DESKTOP-BLT5L3R:~$ more process.c
#include <stdio.h>

int main()
{
    printf("Hello, World!\n");

    return 0;
}
varshini@DESKTOP-BLT5L3R:~$ less hello.c
```

```
#include <stdio.h>
```

```
int main()
```

{

```
printf("Hello, World!\n");
```

```
return 0;
```

}

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C

(END)

```
varshini@DESKTOP-BLT5L3R:~$ wc hello.c
 8 10 81 hello.c
varshini@DESKTOP-BLT5L3R:~$ echo "Hello Linux" > file1.txt
echo "Hello Ubuntu" > file2.txt
varshini@DESKTOP-BLT5L3R:~$ diff file1.txt file2.txt
1c1
< Hello Linux
---
> Hello Ubuntu
varshini@DESKTOP-BLT5L3R:~$ sort file1.txt
Hello Linux
varshini@DESKTOP-BLT5L3R:~$ uniq file2.txt
Hello Ubuntu
varshini@DESKTOP-BLT5L3R:~$ grep main hello.c
int main()
varshini@DESKTOP-BLT5L3R:~$ nano hello.c|
```

```
GNU nano 7.2 hello.c
#include <stdio.h>

int main()
{
    printf("Hello, World!\n");

    return 0;
}
```


GCC(1)

GNU

GCC(1)

NAME

gcc - GNU project C and C++ compiler

SYNOPSIS

```
gcc [-c|-S|-E] [-std=standard]
    [-g] [-pg] [-Olevel]
    [-Wwarn...] [-Wpedantic]
    [-Idir...] [-Ldir...]
    [-Dmacro[=defn]...] [-Umacro]
    [-foption...] [-mmachine-option...]
    [-o outfile] [@file] infile...
```

Only the most useful options are listed here; see below for the remainder. g++ accepts mostly the same options as gcc.

DESCRIPTION

When you invoke GCC, it normally does preprocessing, compilation, assembly and linking. The "overall options" allow you to stop this process at an intermediate stage. For example, the -c option says not to run the linker. Then the output consists of object files output by the assembler.

Other options are passed on to one or more stages of processing. Some options control the preprocessor and others the compiler itself. Yet other options control the assembler and linker; most of these are not documented here, since you rarely need to use any of them.

Most of the command-line options that you can use with GCC are useful for C programs; when an option is only useful with another language (usually C++), the explanation says so explicitly. If the description for a particular option does not mention a source language, you can use that option with all supported languages.

```
varshini@DESKTOP-BLT5L3R:~$ ps
  PID TTY          TIME CMD
  318 pts/0    00:00:00 bash
  761 pts/0    00:00:00 ps
varshini@DESKTOP-BLT5L3R:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1        0  0 13:49 ?        00:00:01 /sbin/init
root           2        1  0 13:49 hvc0     00:00:00 /init
root           8        2  0 13:49 hvc0     00:00:00 plan9 --control-socket 7
root          57        1  0 13:49 ?        00:00:00 /usr/lib/systemd/systemd
root         105        1  0 13:49 ?        00:00:00 /usr/lib/systemd/systemd
systemd+     170        1  0 13:49 ?        00:00:00 /usr/lib/systemd/systemd
systemd+     171        1  0 13:49 ?        00:00:00 /usr/lib/systemd/systemd
root         183        1  0 13:49 ?        00:00:00 /usr/sbin/cron -f -P
message+     184        1  0 13:49 ?        00:00:00 @dbus-daemon --system --
root         191        1  0 13:49 ?        00:00:00 /usr/lib/systemd/systemd
root         193        1  0 13:49 ?        00:00:00 /usr/libexec/wsl-pro-ser
root         203        1  0 13:49 tty1     00:00:00 /sbin/agetty -o -p -- \u
syslog       213        1  0 13:49 ?        00:00:00 /usr/sbin/rsyslogd -n -i
root         231        1  0 13:49 ?        00:00:00 /usr/bin/python3 /usr/sh
root         315        2  0 13:49 ?        00:00:00 /init
root         317       315  0 13:49 ?        00:00:00 /init
varshini     318       317  0 13:49 pts/0    00:00:00 -bash
root         319        2  0 13:49 pts/1    00:00:00 /bin/login -f
varshini     366        1  0 13:49 ?        00:00:00 /usr/lib/systemd/systemd
varshini     368       366  0 13:49 ?        00:00:00 (sd-pam)
varshini     394       319  0 13:49 pts/1    00:00:00 -bash
root         762       105  0 14:22 ?        00:00:00 (udev-worker)
root         763       105  0 14:22 ?        00:00:00 (udev-worker)
varshini     764       318  0 14:23 pts/0    00:00:00 ps -ef
varshini@DESKTOP-BLT5L3R:~$ |
```

```
varshini@DESKTOP-BLT5L3R:~$ top
top - 14:23:29 up 34 min, 1 user, load average: 0.08, 0.02, 0.01
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.3 id, 0.7 wa, 0.0 hi, 0.0 si, 0.
MiB Mem : 7784.6 total, 7167.0 free, 512.8 used, 254.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7271.8 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+
1	root	20	0	21680	13188	9792	S	0.0	0.2	0:01.11
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.02
8	root	20	0	3196	2128	2008	S	0.0	0.0	0:00.00
57	root	19	-1	50444	16200	15060	S	0.0	0.2	0:00.38
105	root	20	0	25204	6568	5132	S	0.0	0.1	0:00.74
170	systemd+	20	0	21344	13184	11056	S	0.0	0.2	0:00.18
171	systemd+	20	0	91036	7964	6996	S	0.0	0.1	0:00.18
183	root	20	0	4244	2680	2432	S	0.0	0.0	0:00.02
184	message+	20	0	9636	5396	4696	S	0.0	0.1	0:00.13
191	root	20	0	17980	8888	7848	S	0.0	0.1	0:00.25
193	root	20	0	1756104	13232	10804	S	0.0	0.2	0:00.30
203	root	20	0	3124	1968	1836	S	0.0	0.0	0:00.02
213	syslog	20	0	222516	5976	4604	S	0.0	0.1	0:00.23
231	root	20	0	107032	23304	13848	S	0.0	0.3	0:00.36
315	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00
317	root	20	0	3200	1252	1108	S	0.0	0.0	0:00.28
318	varshini	20	0	6076	5396	3676	S	0.0	0.1	0:00.30
319	root	20	0	6700	4668	3884	S	0.0	0.1	0:00.03
366	varshini	20	0	20328	11516	9388	S	0.0	0.1	0:00.17
368	varshini	20	0	21168	3576	1848	S	0.0	0.0	0:00.00
394	varshini	20	0	6080	5324	3660	S	0.0	0.1	0:00.04
767	varshini	20	0	9300	5412	3232	R	0.0	0.1	0:00.00

```
varshini@DESKTOP-BLT5L3R:~$ sleep 1000 &
[1] 857
varshini@DESKTOP-BLT5L3R:~$ kill 857
varshini@DESKTOP-BLT5L3R:~$
[1]+  Terminated                  sleep 1000
varshini@DESKTOP-BLT5L3R:~$ sleep 1000 &
[1] 860
varshini@DESKTOP-BLT5L3R:~$ kill -9 860
varshini@DESKTOP-BLT5L3R:~$
[1]+  Killed                        sleep 1000
varshini@DESKTOP-BLT5L3R:~$ |
```

```
varshini@DESKTOP-BLT5L3R:~$ jobs
[1]+  Stopped                      sleep 1000
varshini@DESKTOP-BLT5L3R:~$ bg %1
[1]+ sleep 1000 &
varshini@DESKTOP-BLT5L3R:~$ fg %1
sleep 1000
^C
varshini@DESKTOP-BLT5L3R:~$ ./hello &
[1] 920
```

```

varshini@DESKTOP-BLT5L3R:~$ chmod +x script.sh
chmod: cannot access 'script.sh': No such file or directory
varshini@DESKTOP-BLT5L3R:~$ ls
OSSP      file2.txt  fork_exec.c  hello.c  process.c
file1.txt  fork_exec  hello        myfolder
varshini@DESKTOP-BLT5L3R:~$ find ~ -name "script.sh"
varshini@DESKTOP-BLT5L3R:~$ nano script.sh
varshini@DESKTOP-BLT5L3R:~$ chmod +x script.sh
varshini@DESKTOP-BLT5L3R:~$ ./script.sh
Hello Varshini
varshini@DESKTOP-BLT5L3R:~$ chmod 755 script.sh

```

```

varshini@DESKTOP-BLT5L3R:~$ ln file1.txt file2.txt
ln: failed to create hard link 'file2.txt': File exists
varshini@DESKTOP-BLT5L3R:~$ ln -s file1 shortcut
varshini@DESKTOP-BLT5L3R:~$ df -h

```

Filesystem	Size	Used	Avail	Use%	Mounted on
none	3.9G	0	3.9G	0%	/usr/lib/modules/6.18.33.2-microsoft-s
standard-WSL2					
none	3.9G	4.0K	3.9G	1%	/mnt/wsl
drivers	257G	151G	107G	59%	/usr/lib/wsl/drivers
/dev/sdd	1007G	2.3G	954G	1%	/
none	3.9G	36K	3.9G	1%	/mnt/wslg
none	3.9G	0	3.9G	0%	/usr/lib/wsl/lib
rootfs	3.8G	2.8M	3.8G	1%	/init
none	3.9G	512K	3.9G	1%	/run
none	3.9G	0	3.9G	0%	/run/lock
none	3.9G	0	3.9G	0%	/run/shm
none	3.9G	80K	3.9G	1%	/mnt/wslg/versions.txt
none	3.9G	80K	3.9G	1%	/mnt/wslg/doc
C:\	257G	151G	107G	59%	/mnt/c
D:\	100G	164M	100G	1%	/mnt/d
E:\	100G	822M	100G	1%	/mnt/e
tmpfs	779M	20K	779M	1%	/run/user/1000

```

varshini@DESKTOP-BLT5L3R:~$ |

```

```

varshini@DESKTOP-BLT5L3R:~$ du -sh
304K .
varshini@DESKTOP-BLT5L3R:~$ free -h

```

	total	used	free	shared	buff/cache	available
Mem:	7.6Gi	509Mi	7.1Gi	3.5Mi	201Mi	7.1Gi
Swap:	2.0Gi	0B	2.0Gi			

```

varshini@DESKTOP-BLT5L3R:~$ id
uid=1000(varshini) gid=1000(varshini) groups=1000(varshini),27(sudo),100(users)

```

```
varshini@DESKTOP-BLT5L3R:~$ export PATH=$PATH:..
varshini@DESKTOP-BLT5L3R:~$ alias ll='ls -l'
varshini@DESKTOP-BLT5L3R:~$
```